## CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 5: **Parallel and Intersecting Lines** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The angle between the bisectors of the two angles of a linear pair is: 1

**(a)** $60^\circ$ **(b)** $90^\circ$ **(c)** $180^\circ$ **(d)** $45^\circ$

**2.** Two supplementary angles are in the ratio $1 : 2$. The larger angle is: 1

**(a)** $110^\circ$ **(b)** $130^\circ$ **(c)** $140^\circ$ **(d)** $120^\circ$

**3.** The supplement of the complement of $20^\circ$ is: 1

**(a)** $100^\circ$ **(b)** $110^\circ$ **(c)** $120^\circ$ **(d)** $130^\circ$

**4.** The reflex angle that goes with an angle of $70^\circ$ (so that together they make a complete turn) is: 1

**(a)** $300^\circ$ **(b)** $280^\circ$ **(c)** $290^\circ$ **(d)** $310^\circ$

**5.** Two complementary angles differ by $20^\circ$. The larger angle is: 1

**(a)** $65^\circ$ **(b)** $45^\circ$ **(c)** $75^\circ$ **(d)** $55^\circ$

**6.** Two angles of a linear pair differ by $20^\circ$. The smaller angle is: 1

**(a)** $80^\circ$ **(b)** $90^\circ$ **(c)** $100^\circ$ **(d)** $70^\circ$

**7.** One angle of a linear pair is 2 times the other. The larger angle is: 1

**(a)** $140^\circ$ **(b)** $130^\circ$ **(c)** $110^\circ$ **(d)** $120^\circ$

**8.** If $l \parallel m$, the bisectors of a pair of alternate interior angles are: 1

**(a)** intersecting **(b)** parallel **(c)** coincident **(d)** perpendicular

**9.** **Assertion (A):** The complement of $35^\circ$ is $55^\circ$. 1
**Reason (R):** Two angles are complementary if their sum is $90^\circ$.
Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**10.** **Assertion (A):** The two angles of a linear pair are supplementary. 1
**Reason (R):** Vertically opposite angles are equal.
Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Find the angle whose supplement is $3$ times its complement. **2**

**12.** An angle is $6^\circ$ more than 3 times its complement. Find the angle. **2**

**13.** Two supplementary angles differ by $18^\circ$. Find both angles and the complement of the smaller one. **2**

**14.** Find the angle whose supplement is $4$ times its complement. **2**

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Angles on one side of a straight line at a point are $35^\circ$, $45^\circ$, $50^\circ$ and a fourth angle. Find the fourth angle and name the kind of each angle. **3**

**16.** Two supplementary angles are in the ratio $1 : 4$. Find the complement of the smaller angle and the supplement of the larger angle. **3**

**17.** Two adjacent angles form a linear pair and one of them is $40^\circ$. Find the other angle, the half of each angle and the angle between the two bisectors. **3**

**OR**

Angles on one side of a straight line at a point are $20^\circ$, $75^\circ$, $30^\circ$ and a fourth angle. Find the fourth angle and name the kind of each angle. **3**

**18.** Two supplementary angles are in the ratio $1 : 8$. Find the complement of the smaller angle and the supplement of the larger angle. **3**

**OR**

Two adjacent angles form a linear pair and one of them is $70^\circ$. Find the other angle, the half of each angle and the angle between the two bisectors. **3**

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** Two railway tracks are parallel. A road crosses them and acts as a transversal. At the first crossing, one of the angles is $65^\circ$. **5**

(i) What is the corresponding angle at the second crossing?

(ii) What is the alternate interior angle?

(iii) What is the angle on the same side of the transversal, between the tracks (same-side interior angle)?

(iv) Of the eight angles formed, how many measure $65^\circ$ and how many measure $115^\circ$?

**20.** Three angles on one side of a straight line at a point are in the ratio $2 : 3 : 4$. Find the three angles and name the kind of each. **5**

**OR**

A transversal cuts two parallel lines and two interior angles on the same side are in the ratio $1 : 4$. Find these two angles and say how many of the eight angles measure $36^\circ$ and how many $144^\circ$. **5**

## END OF QUESTION PAPER — Total: 40 Marks